

Lab Sheet # L5

Instructions:

- Submit *.sql file, named as Lab05_6XXXXXX.sql to LAs or instructors prior to My Courses.
- If students cannot finish within the class time, students can submit until the end of the next class.

Self-study:

1. Database: <https://www.mysqltutorial.org/mysql-basics/mysql-drop-database/>
2. Attributes: https://www.mysqltutorial.org/mysql-basics/mysql-auto_increment/
<https://www.mysqltutorial.org/mysql-basics/mysql-generated-columns/>
3. Table: <https://www.mysqltutorial.org/mysql-basics/mysql-temporary-table/>
<https://www.mysqltutorial.org/mysql-basics/mysql-rename-table/>

Today's Task

0. Database Setup: (don't need to submit this task)
 - a. Import the Chinook database by executing the **L4_Chinook.sql** file (from Lect.4).
 - b. If you have already imported the database, it is recommended to drop the database first and then re-import it again.
1. Copy Table: Write a query to create a copy of the **customer** table as **customer_copy** including all data.

```
-- Check row count matches
SELECT
  (SELECT COUNT(*) FROM customer) as original_count,
  (SELECT COUNT(*) FROM customer_copy) as copy_count,
  CASE
    WHEN (SELECT COUNT(*) FROM customer) = (SELECT COUNT(*) FROM customer_copy)
    THEN 'Match'
    ELSE 'Mismatch'
  END as count_check;
```

original_count	copy_count	count_check
59	59	Match

2. Copy the **invoice** table into a new table named **invoice_backup**, including all data.

original_count	backup_count	count_check
412	412	Match

3. Create a copy of the **track** table, but only copy the structure (no data) into a table named **tracks_structure_only**.

(Hint: Copy nothing from All, or use LIKE)

TrackId	Name	AlbumId	MediaTypeId	GenreId	Composer	Milliseconds	Bytes	UnitPrice
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4. Create a copy of the **customer** table where only customers from "Canada" are included. Name the new table **customer_canada**.

```
-- Check all records are from Canada
• SELECT
    COUNT(*) as total_records,
    SUM(CASE WHEN Country = 'Canada' THEN 1 ELSE 0 END) as canadian_records,
    CASE
        WHEN COUNT(*) = SUM(CASE WHEN Country = 'Canada' THEN 1 ELSE 0 END)
        THEN 'All Records Canadian'
        ELSE 'Error: Non-Canadian Records Found'
    END as verification_status
FROM customer_canada;
```

total_records	canadian_records	verification_status
8	8	All Records Canadian

5. Add an AUTO_INCREMENT property to the **CustomerId** column in the **customer_copy** table.

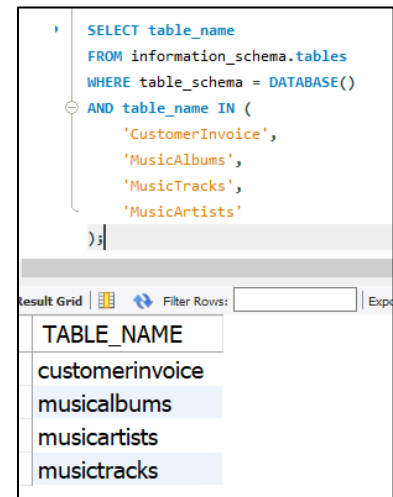
Field	Type	Null	Key	Default	Extra
CustomerId	int	NO	PRI	NULL	auto_increment
FirstName	varchar(40)	NO		NULL	
LastName	varchar(20)	NO		NULL	
Company	varchar(80)	YES		NULL	

6. Try to add some new customers to the **customer_copy** table and verify the AUTO_INCREMENT functionality.

CustomerId	FirstName	LastName	Company	Address	City	State	Country	PostalCode	Phone
61	Emma	Johnson	NULL	NULL	NULL	NULL	Canada	NULL	+1-553
60	John	Smith	NULL	NULL	NULL	NULL	USA	NULL	+1-553
59	Puja	Srivastava	NULL	3,Raj Bhavan Road	Bangalore	NULL	India	560001	+91 08
58	Manoj	Pareek	NULL	12,Community Centre	Delhi	NULL	India	110017	+91 01
57	Luis	Rojas	NULL	Calle Lira, 198	Santiago	NULL	Chile	NULL	+56 (0

7. Rename Table

- Rename the **invoice** table to **CustomerInvoice**.
- Rename the **album** table to **MusicAlbums**.
- Rename the **track** table to **MusicTracks**.
- Rename the **artist** table to **MusicArtists**.



- Use Generated Column: Add a generated column to the **track** table to calculate the track's length in minutes. The generated column should be named **LengthInMinutes**.

TrackId	Name	AlbumId	MediaTypeId	GenreId	Composer	Milliseconds	Bytes	UnitPrice	LengthInMinutes
1	For Those About To Rock ...	1	1	1	Angus Young, Malcolm Yo...	343719	11170334	0.99	5.73
2	Balls to the Wall	2	2	1	U. Dirkschneider, W. Hoff...	342562	5510424	0.99	5.71
3	Fast As a Shark	3	2	1	F. Baltes, S. Kaufman, U. ...	230619	3990994	0.99	3.84
4	Restless and Wild	3	2	1	F. Baltes, R.A. Smith-Dies...	252051	4331779	0.99	4.20
5	Princess of the Dawn	3	2	1	Deaffy & R.A. Smith-Diesel	375418	6290521	0.99	6.26
6	Put The Finger On You	1	1	1	Angus Young, Malcolm Yo...	205662	6713451	0.99	3.43